



Mathematics: AA 1.1 Arithmetic and geometric sequences and series State the formula for the nth term of a geometric sequence and explain each symbol. _____ IB-MATHAA-1.1-1 CONCEPT	Mathematics: AA 1.1 Arithmetic and geometric sequences and series Why does the sum to infinity formula $S_{\infty} = u_1/(1-r)$ only apply for certain r values? _____ IB-MATHAA-1.1-2 CONCEPT	Mathematics: AA 1.1 Arithmetic and geometric sequences and series An AP has $u_1=2$, $d=5$. What is u_{10} ? A 45 B 47 C 50 D 52 _____ IB-MATHAA-1.1-3 MULTIPLE CHOICE
Mathematics: AA 1.1 Arithmetic and geometric sequences and series Find the sum of the first 6 terms of the GP 3,6,12,... _____ IB-MATHAA-1.1-4 PRACTICE	Mathematics: AA 1.2 Exponents and logarithms State the change of base formula for $\log_a(x)$ in terms of natural logarithms. _____ IB-MATHAA-1.2-1 CONCEPT	Mathematics: AA 1.2 Exponents and logarithms Why must the domain be checked after solving an equation involving $\log(x-2)$? _____ IB-MATHAA-1.2-2 CONCEPT
Mathematics: AA 1.2 Exponents and logarithms $\log_5(125)$ equals: A 2 B 3 C 5 D 25 _____ IB-MATHAA-1.2-3 MULTIPLE CHOICE	Mathematics: AA 1.2 Exponents and logarithms Solve $3^x = 50$ giving x to 3 s.f. _____ IB-MATHAA-1.2-4 PRACTICE	Mathematics: AA 1.3 Financial applications of geometric sequences Write the compound interest formula and define each symbol. _____ IB-MATHAA-1.3-1 CONCEPT
Mathematics: AA 1.3 Financial applications of geometric sequences Why is compound growth modelled by a geometric, not arithmetic, sequence? _____ IB-MATHAA-1.3-2 CONCEPT	Mathematics: AA 1.3 Financial applications of geometric sequences \$1000 grows at 10% p.a. compounded annually. Value after 3 years? A \$1300 B \$1330 C \$1331 D \$1210 _____ IB-MATHAA-1.3-3 MULTIPLE CHOICE	Mathematics: AA 1.3 Financial applications of geometric sequences Find the value of \$800 depreciating at 20% per year after 3 years. _____ IB-MATHAA-1.3-4 PRACTICE
Mathematics: AA 1.4 The binomial theorem State the binomial theorem formula for $(a+b)^n$. _____ IB-MATHAA-1.4-1 CONCEPT	Mathematics: AA 1.4 The binomial theorem How do you find a specific term (not the whole expansion) using the binomial theorem? _____ IB-MATHAA-1.4-2 CONCEPT	Mathematics: AA 1.4 The binomial theorem The coefficient of x^2 in $(1+x)^4$ is: A 4 B 6 C 8 D 12 _____ IB-MATHAA-1.4-3 MULTIPLE CHOICE
Mathematics: AA 1.4 The binomial theorem Find the coefficient of x^2 in $(2-x)^4$. _____ IB-MATHAA-1.4-4 PRACTICE	Mathematics: AA 1.5 Complex numbers (HL) State De Moivre's theorem and what it is used for. _____ IB-MATHAA-1.5-1 CONCEPT	Mathematics: AA 1.5 Complex numbers (HL) How do you find the modulus and argument of a complex number $a+bi$, and what care is needed for the argument? _____ IB-MATHAA-1.5-2 CONCEPT



Mathematics: AA 1.5 Complex numbers (HL) (2+i)(1-i) equals: A 1-i B 3-i C 1+3i D 3+i IB-MATHAA-1.5-3MULTIPLE CHOICE	Mathematics: AA 1.5 Complex numbers (HL) Find $z_1 \times z_2$ in polar form given $z_1 = 2(\cos 30^\circ + i \sin 30^\circ)$, $z_2 = 3(\cos 40^\circ + i \sin 40^\circ)$. IB-MATHAA-1.5-4PRACTICE	Mathematics: AA 1.6 Proof by induction (HL) List the three components required in a complete proof by induction. IB-MATHAA-1.6-1CONCEPT
Mathematics: AA 1.6 Proof by induction (HL) Why is it insufficient to only check the base case and a few values of n? IB-MATHAA-1.6-2CONCEPT	Mathematics: AA 1.6 Proof by induction (HL) Induction is most suitable for proving statements about: A A single real number B All positive integers n C All real numbers x D Continuous functions only IB-MATHAA-1.6-3MULTIPLE CHOICE	Mathematics: AA 1.6 Proof by induction (HL) State the base case check needed to prove $2^n > n$ for all $n \geq 1$. IB-MATHAA-1.6-4PRACTICE
Mathematics: AA 1.7 Systems of linear equations (HL) Name the three possible solution types for a 3×3 linear system and their geometric meaning. IB-MATHAA-1.7-1CONCEPT	Mathematics: AA 1.7 Systems of linear equations (HL) What row operation is used to eliminate a variable between two equations in Gaussian elimination? IB-MATHAA-1.7-2CONCEPT	Mathematics: AA 1.7 Systems of linear equations (HL) Which row operation does NOT change the solution set of a linear system? A Multiplying a row by 0 B Swapping two rows C Deleting a row D Adding a constant to one side only IB-MATHAA-1.7-3MULTIPLE CHOICE
Mathematics: AA 1.7 Systems of linear equations (HL) Solve $x+y=5$, $2x-y=1$. IB-MATHAA-1.7-4PRACTICE	Mathematics: AA 2.1 Functions, domain and range What test determines whether a graph represents a function, and what does it check? IB-MATHAA-2.1-1CONCEPT	Mathematics: AA 2.1 Functions, domain and range Give two common reasons a value must be excluded from a function's domain. IB-MATHAA-2.1-2CONCEPT
Mathematics: AA 2.1 Functions, domain and range The range of $f(x)=x^2$ for domain $x \in \mathbb{R}$ is: A $f(x) \geq 0$ B $f(x) > 0$ C All real numbers D $f(x) \leq 0$ IB-MATHAA-2.1-3MULTIPLE CHOICE	Mathematics: AA 2.1 Functions, domain and range State the domain of $f(x)=\sqrt{x+4}$. IB-MATHAA-2.1-4PRACTICE	Mathematics: AA 2.2 Composite and inverse functions What is the geometric relationship between the graphs of $y=f(x)$ and $y=f^{-1}(x)$? IB-MATHAA-2.2-1CONCEPT
Mathematics: AA 2.2 Composite and inverse functions Why does a function need to be one-to-one (a bijection) for its inverse to exist as a function? IB-MATHAA-2.2-2CONCEPT	Mathematics: AA 2.2 Composite and inverse functions If $f(x)=x+5$, then $f^{-1}(x)$ is: A $x+5$ B $x-5$ C $1/(x+5)$ D $-x-5$ IB-MATHAA-2.2-3MULTIPLE CHOICE	Mathematics: AA 2.2 Composite and inverse functions If $f(x)=2x$ and $g(x)=x+3$, find $(f \circ g)(1)$. IB-MATHAA-2.2-4PRACTICE



Mathematics: AA 2.3 Quadratic functions and the discriminant	Mathematics: AA 2.3 Quadratic functions and the discriminant	Mathematics: AA 2.3 Quadratic functions and the discriminant
State the discriminant formula and what each sign of Δ tells you about the roots of $ax^2+bx+c=0$.	How do you find the vertex of a quadratic directly from its equation without completing the square?	The equation $x^2-4x+5=0$ has: A Two distinct real roots B One repeated real root C No real roots D Three real roots
IB-MATHAA-2.3-1CONCEPT	IB-MATHAA-2.3-2CONCEPT	IB-MATHAA-2.3-3MULTIPLE CHOICE
Mathematics: AA 2.3 Quadratic functions and the discriminant	Mathematics: AA 2.4 Transformations of graphs	Mathematics: AA 2.4 Transformations of graphs
Find the vertex of $f(x)=x^2-2x+4$.	Describe the effect of the transformation $y=f(x)+d$ and $y=f(x+c)$ on the graph of $y=f(x)$.	What is the effect of a on the graph of $y=af(x)$ when $-1<a<0$?
IB-MATHAA-2.3-4PRACTICE	IB-MATHAA-2.4-1CONCEPT	IB-MATHAA-2.4-2CONCEPT
Mathematics: AA 2.4 Transformations of graphs	Mathematics: AA 2.4 Transformations of graphs	Mathematics: AA 2.5 Exponential and logarithmic functions
$y=3f(x)$ transforms $y=f(x)$ by: A Horizontal stretch factor 3 B Vertical stretch factor 3 C Vertical translation of 3 D Horizontal translation of 3	If $y=f(x)$ has a maximum at (1,5), find the maximum point of $y=f(x)-2$.	State the domain, range, and asymptote of $y=\ln(x)$.
IB-MATHAA-2.4-3MULTIPLE CHOICE	IB-MATHAA-2.4-4PRACTICE	IB-MATHAA-2.5-1CONCEPT
Mathematics: AA 2.5 Exponential and logarithmic functions	Mathematics: AA 2.5 Exponential and logarithmic functions	Mathematics: AA 2.5 Exponential and logarithmic functions
Why are $y=e^x$ and $y=\ln(x)$ reflections of each other in the line $y=x$?	The y-intercept of $y=4^x + 1$ is: A 0 B 1 C 2 D 4	State the horizontal asymptote of $y=7(0.5)^x - 4$.
IB-MATHAA-2.5-2CONCEPT	IB-MATHAA-2.5-3MULTIPLE CHOICE	IB-MATHAA-2.5-4PRACTICE
Mathematics: AA 2.6 Rational functions (HL)	Mathematics: AA 2.6 Rational functions (HL)	Mathematics: AA 2.6 Rational functions (HL)
For $f(x)=(ax+b)/(cx+d)$, state how to find the vertical and horizontal asymptotes.	Why does a rational function $(ax+b)/(cx+d)$ never actually reach its horizontal asymptote?	The vertical asymptote of $f(x)=(x+2)/(2x-6)$ is: A $x=6$ B $x=3$ C $x=-2$ D $x=2$
IB-MATHAA-2.6-1CONCEPT	IB-MATHAA-2.6-2CONCEPT	IB-MATHAA-2.6-3MULTIPLE CHOICE
Mathematics: AA 2.6 Rational functions (HL)	Mathematics: AA 2.7 Polynomial functions and factor theorem (HL)	Mathematics: AA 2.7 Polynomial functions and factor theorem (HL)
State the horizontal asymptote of $f(x)=(4x-1)/(2x+5)$.	State the factor theorem.	How does the remainder theorem differ from the factor theorem?
IB-MATHAA-2.6-4PRACTICE	IB-MATHAA-2.7-1CONCEPT	IB-MATHAA-2.7-2CONCEPT
Mathematics: AA 2.7 Polynomial functions and factor theorem (HL)	Mathematics: AA 2.7 Polynomial functions and factor theorem (HL)	Mathematics: AA 3.1 Volume and surface area of 3D solids
$P(x)=x^3-x-6$ has $P(2)=0$. Which is a factor of $P(x)$? A $(x-6)$ B $(x+2)$ C $(x-2)$ D $(x+6)$	Find the remainder when x^3-2x+1 is divided by $(x-2)$.	State the volume formula for a cone and for a sphere.
IB-MATHAA-2.7-3MULTIPLE CHOICE	IB-MATHAA-2.7-4PRACTICE	IB-MATHAA-3.1-1CONCEPT



Mathematics: AA 3.1 Volume and surface area of 3D solids	Mathematics: AA 3.1 Volume and surface area of 3D solids	Mathematics: AA 3.1 Volume and surface area of 3D solids
What is the slant height of a cone and how is it calculated?	The curved surface area of a cylinder with $r=2$, $h=7$ is: A 14π B 28π C 44π D 56π	Find the volume of a cuboid with dimensions 3 cm $\times$ 4 cm $\times$ 5 cm.
IB-MATHAA-3.1-2 CONCEPT	IB-MATHAA-3.1-3 MULTIPLE CHOICE	IB-MATHAA-3.1-4 PRACTICE
Mathematics: AA 3.2 Right-angled trigonometry and bearings	Mathematics: AA 3.2 Right-angled trigonometry and bearings	Mathematics: AA 3.2 Right-angled trigonometry and bearings
State the SOH-CAH-TOA ratios for a right-angled triangle.	How are bearings measured, and what is the rule for expressing them?	The angle of elevation and angle of depression between two points are: A Always different B Equal (alternate angles) C Complementary D Supplementary
IB-MATHAA-3.2-1 CONCEPT	IB-MATHAA-3.2-2 CONCEPT	IB-MATHAA-3.2-3 MULTIPLE CHOICE
Mathematics: AA 3.2 Right-angled trigonometry and bearings	Mathematics: AA 3.3 Non-right triangles: sine and cosine rules	Mathematics: AA 3.3 Non-right triangles: sine and cosine rules
A ladder of length 6 m leans against a wall making 60° with the ground. How high up the wall does it reach?	State the sine rule and the cosine rule (side form).	When does the 'ambiguous case' arise when using the sine rule, and how do you check for it?
IB-MATHAA-3.2-4 PRACTICE	IB-MATHAA-3.3-1 CONCEPT	IB-MATHAA-3.3-2 CONCEPT
Mathematics: AA 3.3 Non-right triangles: sine and cosine rules	Mathematics: AA 3.3 Non-right triangles: sine and cosine rules	Mathematics: AA 3.4 The unit circle and radian measure
The area of a triangle with sides 6 and 8 and included angle 90° is: A 24 B 48 C 14 D 30	In triangle ABC, $a=6$, $b=6$, $C=60^\circ$. Find c.	State the formulas for arc length and sector area in terms of radius r and angle θ in radians.
IB-MATHAA-3.3-3 MULTIPLE CHOICE	IB-MATHAA-3.3-4 PRACTICE	IB-MATHAA-3.4-1 CONCEPT
Mathematics: AA 3.4 The unit circle and radian measure	Mathematics: AA 3.4 The unit circle and radian measure	Mathematics: AA 3.4 The unit circle and radian measure
On the unit circle, how are $\cos\theta$ and $\sin\theta$ defined geometrically?	$\pi/3$ radians is equivalent to: A 30° B 45° C 60° D 90°	Find the arc length of a sector with radius 5 cm and angle $\pi/2$ radians.
IB-MATHAA-3.4-2 CONCEPT	IB-MATHAA-3.4-3 MULTIPLE CHOICE	IB-MATHAA-3.4-4 PRACTICE
Mathematics: AA 3.5 Trigonometric identities and equations	Mathematics: AA 3.5 Trigonometric identities and equations	Mathematics: AA 3.5 Trigonometric identities and equations
State the Pythagorean identity and one double angle formula for cosine.	When solving a trig equation like $\sin(2x)=k$ over a domain, why must you find ALL solutions of the argument before dividing back by 2?	$\sin^2x + \cos^2x$ is always equal to: A 0 B 1 C $2\sin x \cos x$ D $\tan^2x$
IB-MATHAA-3.5-1 CONCEPT	IB-MATHAA-3.5-2 CONCEPT	IB-MATHAA-3.5-3 MULTIPLE CHOICE



Mathematics: AA 3.5 Trigonometric identities and equations	Mathematics: AA 3.6 Vectors and scalar/vector products (HL)	Mathematics: AA 3.6 Vectors and scalar/vector products (HL)
Solve $\sin x = 1/2$ for $0 \leq x \leq 2\pi$.	State the formula for the scalar (dot) product of two vectors in terms of their magnitudes and the angle between them.	What geometric quantity does $ a \times b $ represent, and how is it used to find a triangle's area?
IB-MATHAA-3.5-4PRACTICE	IB-MATHAA-3.6-1CONCEPT	IB-MATHAA-3.6-2CONCEPT
Mathematics: AA 3.6 Vectors and scalar/vector products (HL)	Mathematics: AA 3.6 Vectors and scalar/vector products (HL)	Mathematics: AA 3.7 Lines and planes in three dimensions (HL)
For $a=(1,0,0)$ and $b=(0,1,0)$, $a \cdot b$ equals: A 1 B 0 C -1 D (0,0,1)	Find $ v $ for $v=(3,4,0)$.	Write the vector equation of a line and the equation of a plane, defining all symbols.
IB-MATHAA-3.6-3MULTIPLE CHOICE	IB-MATHAA-3.6-4PRACTICE	IB-MATHAA-3.7-1CONCEPT
Mathematics: AA 3.7 Lines and planes in three dimensions (HL)	Mathematics: AA 3.7 Lines and planes in three dimensions (HL)	Mathematics: AA 3.7 Lines and planes in three dimensions (HL)
How do you find the angle between a line and a plane, given the line's direction vector and the plane's normal vector?	Two planes with normals $n_1=(1,0,0)$ and $n_2=(0,1,0)$ are: A Parallel B Perpendicular C Identical D Intersecting at an obtuse angle	Find the point where the line $r=(0,0,0)+t(1,1,1)$ meets the plane $x+y+z=6$.
IB-MATHAA-3.7-2CONCEPT	IB-MATHAA-3.7-3MULTIPLE CHOICE	IB-MATHAA-3.7-4PRACTICE
Mathematics: AA 4.1 Sampling and data collection	Mathematics: AA 4.1 Sampling and data collection	Mathematics: AA 4.1 Sampling and data collection
Describe stratified sampling and why it is used.	What is sampling bias, and give one cause of it.	Which method gives every member of the population an equal chance of selection? A Convenience sampling B Quota sampling C Simple random sampling D Systematic sampling
IB-MATHAA-4.1-1CONCEPT	IB-MATHAA-4.1-2CONCEPT	IB-MATHAA-4.1-3MULTIPLE CHOICE
Mathematics: AA 4.1 Sampling and data collection	Mathematics: AA 4.2 Statistical measures and displays	Mathematics: AA 4.2 Statistical measures and displays
A population has 400 men and 600 women. Find the number of women in a stratified sample of size 50.	State the formula for the interquartile range and the rule for identifying an outlier using it.	Why is the median often preferred over the mean when a data set contains outliers?
IB-MATHAA-4.1-4PRACTICE	IB-MATHAA-4.2-1CONCEPT	IB-MATHAA-4.2-2CONCEPT
Mathematics: AA 4.2 Statistical measures and displays	Mathematics: AA 4.2 Statistical measures and displays	Mathematics: AA 4.3 Correlation and linear regression
A box plot's central line represents the: A Mean B Mode C Median D Range	Find the range and IQR of the data set: 3, 5, 5, 8, 9, 11, 14.	What does Pearson's correlation coefficient r measure, and what range of values can it take?
IB-MATHAA-4.2-3MULTIPLE CHOICE	IB-MATHAA-4.2-4PRACTICE	IB-MATHAA-4.3-1CONCEPT



Mathematics: AA 4.3 Correlation and linear regression	Mathematics: AA 4.3 Correlation and linear regression	Mathematics: AA 4.3 Correlation and linear regression
Why is extrapolation using a regression line considered unreliable?	Which value of r shows the weakest linear correlation? A $r=0.95$ B $r=-0.88$ C $r=0.05$ D $r=-1.0$	A regression line is $y=3x+2$. Estimate y when $x=4$.
IB-MATHAA-4.3-2 CONCEPT	IB-MATHAA-4.3-3 MULTIPLE CHOICE	IB-MATHAA-4.3-4 PRACTICE
Mathematics: AA 4.4 Probability, Venn and tree diagrams	Mathematics: AA 4.4 Probability, Venn and tree diagrams	Mathematics: AA 4.4 Probability, Venn and tree diagrams
State the addition rule for $P(A \cup B)$ and how it simplifies for mutually exclusive events.	State the condition for two events A and B to be independent, in terms of probabilities.	$P(A)=0.6$, $P(B)=0.3$, $P(A \cap B)=0.18$. Are A and B independent? A Yes, since $0.6 \times 0.3 = 0.18$ B No, since $P(A \cap B) \neq 0$ C Yes, since $P(A \cup B) > 0$ D Cannot be determined
IB-MATHAA-4.4-1 CONCEPT	IB-MATHAA-4.4-2 CONCEPT	IB-MATHAA-4.4-3 MULTIPLE CHOICE
Mathematics: AA 4.4 Probability, Venn and tree diagrams	Mathematics: AA 4.5 Discrete random variables and the binomial distribution	Mathematics: AA 4.5 Discrete random variables and the binomial distribution
$P(A)=0.4$, $P(B)=0.5$, A and B mutually exclusive. Find $P(A \cup B)$.	State the formulas for the mean and variance of a binomial distribution $X \sim B(n, p)$.	What conditions must be satisfied for a random variable to be modelled by a binomial distribution?
IB-MATHAA-4.4-4 PRACTICE	IB-MATHAA-4.5-1 CONCEPT	IB-MATHAA-4.5-2 CONCEPT
Mathematics: AA 4.5 Discrete random variables and the binomial distribution	Mathematics: AA 4.5 Discrete random variables and the binomial distribution	Mathematics: AA 4.6 The normal distribution
For a discrete random variable, $\sum P(X=x)$ over all values of x must equal: A 0 B 0.5 C 1 D n	$X \sim B(5, 0.5)$. Find $E(X)$ and $\text{Var}(X)$.	State the standardisation formula for converting a normal variable X to the standard normal Z .
IB-MATHAA-4.5-3 MULTIPLE CHOICE	IB-MATHAA-4.5-4 PRACTICE	IB-MATHAA-4.6-1 CONCEPT
Mathematics: AA 4.6 The normal distribution	Mathematics: AA 4.6 The normal distribution	Mathematics: AA 4.6 The normal distribution
State the approximate percentages of data within 1, 2, and 3 standard deviations of the mean in a normal distribution.	For $X \sim N(\mu, \sigma^2)$, $P(X < \mu)$ equals: A 0 B 0.25 C 0.5 D 1	$X \sim N(20, 5^2)$. Find the z-score for $x=25$.
IB-MATHAA-4.6-2 CONCEPT	IB-MATHAA-4.6-3 MULTIPLE CHOICE	IB-MATHAA-4.6-4 PRACTICE
Mathematics: AA 4.7 Bayes' theorem (HL)	Mathematics: AA 4.7 Bayes' theorem (HL)	Mathematics: AA 4.7 Bayes' theorem (HL)
State Bayes' theorem for $P(A B)$.	How is the law of total probability used within Bayes' theorem?	In Bayes' theorem, $P(A)$ (before observing B) is called the: A Posterior probability B Prior probability C Conditional probability D Joint probability
IB-MATHAA-4.7-1 CONCEPT	IB-MATHAA-4.7-2 CONCEPT	IB-MATHAA-4.7-3 MULTIPLE CHOICE



Mathematics: AA 4.7 Bayes' theorem (HL) $P(B A)=0.8$, $P(A)=0.25$, $P(B)=0.4$. Find $P(A B)$. IB-MATHAA-4.7-4 PRACTICE	Mathematics: AA 5.1 Limits and the derivative from first principles State the definition of the derivative of $f(x)$ from first principles. IB-MATHAA-5.1-1 CONCEPT	Mathematics: AA 5.1 Limits and the derivative from first principles What does the derivative $f'(x)$ represent geometrically? IB-MATHAA-5.1-2 CONCEPT
Mathematics: AA 5.1 Limits and the derivative from first principles For $f(x)=5$ (a constant), $f'(x)$ from first principles equals: A 5 B 0 C x D Undefined IB-MATHAA-5.1-3 MULTIPLE CHOICE	Mathematics: AA 5.1 Limits and the derivative from first principles Using first principles, find $f'(x)$ for $f(x)=4x$. IB-MATHAA-5.1-4 PRACTICE	Mathematics: AA 5.2 Differentiation rules and tangents/normals State the product rule and the quotient rule for differentiation. IB-MATHAA-5.2-1 CONCEPT
Mathematics: AA 5.2 Differentiation rules and tangents/normals How is the gradient of the normal to a curve at a point related to the gradient of the tangent there? IB-MATHAA-5.2-2 CONCEPT	Mathematics: AA 5.2 Differentiation rules and tangents/normals $d/dx(x^5)$ equals: A x^4 B $5x^6$ C $5x^4$ D $x^6/6$ IB-MATHAA-5.2-3 MULTIPLE CHOICE	Mathematics: AA 5.2 Differentiation rules and tangents/normals Find dy/dx for $y=x^2+3x$ at $x=2$. IB-MATHAA-5.2-4 PRACTICE
Mathematics: AA 5.3 Optimisation and related rates How do you classify a critical point $x=a$ (where $f'(a)=0$) using the second derivative test? IB-MATHAA-5.3-1 CONCEPT	Mathematics: AA 5.3 Optimisation and related rates In a related rates problem, what role does the chain rule play? IB-MATHAA-5.3-2 CONCEPT	Mathematics: AA 5.3 Optimisation and related rates For $f(x)=x^2-6x+5$, the minimum value occurs at: A $x=3$ B $x=6$ C $x=-3$ D $x=0$ IB-MATHAA-5.3-3 MULTIPLE CHOICE
Mathematics: AA 5.3 Optimisation and related rates Find the critical point of $f(x)=x^2-4x+1$ and state whether it is a maximum or minimum. IB-MATHAA-5.3-4 PRACTICE	Mathematics: AA 5.4 Kinematics using calculus State the relationships between displacement, velocity, and acceleration using calculus notation. IB-MATHAA-5.4-1 CONCEPT	Mathematics: AA 5.4 Kinematics using calculus How do you determine when a moving particle is momentarily at rest? IB-MATHAA-5.4-2 CONCEPT
Mathematics: AA 5.4 Kinematics using calculus If $a(t)=4$ (constant) and $v(0)=2$, what is $v(t)$? A $4t$ B $4t+2$ C $2t+4$ D $4t-2$ IB-MATHAA-5.4-3 MULTIPLE CHOICE	Mathematics: AA 5.4 Kinematics using calculus A particle has $s(t)=t^2-4t$. Find its velocity at $t=3$. IB-MATHAA-5.4-4 PRACTICE	Mathematics: AA 5.5 Integration and areas under curves State the power rule for integration and one standard exponential/logarithmic integral. IB-MATHAA-5.5-1 CONCEPT



Mathematics: AA 5.5 Integration and areas under curves Why must you check which curve is above the other before computing the area between two curves? IB-MATHAA-5.5-2 CONCEPT	Mathematics: AA 5.5 Integration and areas under curves $\int [1 \text{ to } 3] 2x \, dx$ equals: A 4 B 6 C 8 D 10 IB-MATHAA-5.5-3 MULTIPLE CHOICE	Mathematics: AA 5.5 Integration and areas under curves Find $\int (3x^2-2)dx$. IB-MATHAA-5.5-4 PRACTICE
Mathematics: AA 5.6 Volumes of revolution (HL) State the formula for the volume of revolution when $y=f(x)$ is rotated 360° about the x-axis between $x=a$ and $x=b$. IB-MATHAA-5.6-1 CONCEPT	Mathematics: AA 5.6 Volumes of revolution (HL) Why does the volume of revolution formula involve y^2 rather than y ? IB-MATHAA-5.6-2 CONCEPT	Mathematics: AA 5.6 Volumes of revolution (HL) The region under $y=2$ (constant) from $x=0$ to $x=5$, rotated about the x-axis, gives volume: A 10π B 20π C 25π D 50π IB-MATHAA-5.6-3 MULTIPLE CHOICE
Mathematics: AA 5.6 Volumes of revolution (HL) Set up (do not evaluate) the integral for the volume when $y=x^2$ ($0 \leq x \leq 1$) is rotated about the x-axis. IB-MATHAA-5.6-4 PRACTICE	Mathematics: AA 5.7 Further differentiation: implicit and higher derivatives (HL) When differentiating an equation implicitly, how do you handle a term like y^2 or xy with respect to x ? IB-MATHAA-5.7-1 CONCEPT	Mathematics: AA 5.7 Further differentiation: implicit and higher derivatives (HL) Why is implicit differentiation necessary for curves like $x^2+y^2=25$? IB-MATHAA-5.7-2 CONCEPT
Mathematics: AA 5.7 Further differentiation: implicit and higher derivatives (HL) For $x+y^2=10$, dy/dx equals: A $-1/(2y)$ B $1/(2y)$ C $-2y$ D $2y$ IB-MATHAA-5.7-3 MULTIPLE CHOICE	Mathematics: AA 5.7 Further differentiation: implicit and higher derivatives (HL) Find dy/dx for the curve $x^2-y=4$. IB-MATHAA-5.7-4 PRACTICE	Mathematics: AA 5.8 Further integration: substitution and by parts (HL) State the integration by parts formula. IB-MATHAA-5.8-1 CONCEPT
Mathematics: AA 5.8 Further integration: substitution and by parts (HL) When using substitution $u=g(x)$, what must you remember to convert besides the function itself? IB-MATHAA-5.8-2 CONCEPT	Mathematics: AA 5.8 Further integration: substitution and by parts (HL) For $\int 3x^2(x^3+1)^5 \, dx$, a suitable substitution is: A $u=x^2$ B $u=x^3+1$ C $u=x^5$ D $u=3x^2$ IB-MATHAA-5.8-3 MULTIPLE CHOICE	Mathematics: AA 5.8 Further integration: substitution and by parts (HL) Find $\int 2x \cdot \cos(x^2)dx$ using substitution. IB-MATHAA-5.8-4 PRACTICE
Mathematics: AA 5.9 Differential equations and Maclaurin series (HL) Outline the method for solving a separable differential equation $dy/dx=f(x)g(y)$. IB-MATHAA-5.9-1 CONCEPT	Mathematics: AA 5.9 Differential equations and Maclaurin series (HL) State the general Maclaurin series formula for $f(x)$ about $x=0$. IB-MATHAA-5.9-2 CONCEPT	Mathematics: AA 5.9 Differential equations and Maclaurin series (HL) For the Maclaurin series of $\sin(x)$, the coefficient of x^3 is: A $1/6$ B $-1/6$ C $1/3$ D $-1/3$ IB-MATHAA-5.9-3 MULTIPLE CHOICE

**Mathematics: AA**

5.9 Differential equations and Maclaurin series (HL)

Solve $\frac{dy}{dx} = 2y$ (separable), given
 $y=5$ when $x=0$.

IB-MATHAA-5.9-4

PRACTICE



Mathematics: AA 1.1 Arithmetic and geometric sequences and series B. $u_{10}=2+9(5)=47$. IB-MATHAA-1.1-3ANSWER	Mathematics: AA 1.1 Arithmetic and geometric sequences and series Only when $ r <1$, since then $r^n \rightarrow 0$ as $n \rightarrow \infty$ so the partial sums converge to a finite limit; otherwise the series diverges. IB-MATHAA-1.1-2ANSWER	Mathematics: AA 1.1 Arithmetic and geometric sequences and series $u_n = u_1 \cdot r^{(n-1)}$, where u_1 is the first term, r the common ratio, and n the term number. IB-MATHAA-1.1-1ANSWER
Mathematics: AA 1.2 Exponents and logarithms Because logarithms are only defined for positive arguments, so any solution making $x-2 \leq 0$ (or the other argument ≤ 0) must be rejected. IB-MATHAA-1.2-2ANSWER	Mathematics: AA 1.2 Exponents and logarithms $\log_a(x) = \ln(x)/\ln(a)$. IB-MATHAA-1.2-1ANSWER	Mathematics: AA 1.1 Arithmetic and geometric sequences and series $r=2$, $S_n = u_1(r^n - 1)/(r - 1) = 3(2^6 - 1)/1 = 3(63) = 189$. IB-MATHAA-1.1-4ANSWER
Mathematics: AA 1.3 Financial applications of geometric sequences $FV = P(1+r/(100k))^n$: P principal, r nominal annual rate (%), k compounding periods per year, n number of years. IB-MATHAA-1.3-1ANSWER	Mathematics: AA 1.2 Exponents and logarithms $x = \ln 50 / \ln 3 = 3.9120 / 1.0986 = 3.56$. IB-MATHAA-1.2-4ANSWER	Mathematics: AA 1.2 Exponents and logarithms B. $5^3 = 125$, so $\log_5(125) = 3$. IB-MATHAA-1.2-3ANSWER
Mathematics: AA 1.3 Financial applications of geometric sequences $800 \times 0.8^3 = 800 \times 0.512 = \409.60 . IB-MATHAA-1.3-4ANSWER	Mathematics: AA 1.3 Financial applications of geometric sequences C. $1000(1.1)^3 = 1331$. IB-MATHAA-1.3-3ANSWER	Mathematics: AA 1.3 Financial applications of geometric sequences Because each compounding period multiplies the current value by a constant factor $(1+r/(100k))$, giving a common ratio rather than a common difference. IB-MATHAA-1.3-2ANSWER
Mathematics: AA 1.4 The binomial theorem B. $4C_2 = 6$. IB-MATHAA-1.4-3ANSWER	Mathematics: AA 1.4 The binomial theorem Use the general term $nCr \cdot a^{(n-r)} \cdot b^r$ and solve for the value of r that gives the required power of the variable, then substitute. IB-MATHAA-1.4-2ANSWER	Mathematics: AA 1.4 The binomial theorem $(a+b)^n = \sum_{r=0}^n nCr \cdot a^{(n-r)} \cdot b^r$. IB-MATHAA-1.4-1ANSWER
Mathematics: AA 1.5 Complex numbers (HL) $r = \sqrt{a^2 + b^2}$; $\theta = \arctan(b/a)$ adjusted for the correct quadrant based on the signs of a and b , since $\arctan$ alone only gives values in $(-\pi/2, \pi/2)$. IB-MATHAA-1.5-2ANSWER	Mathematics: AA 1.5 Complex numbers (HL) $z^n = r^n(\cos n\theta + i \sin n\theta)$ for $z = r(\cos \theta + i \sin \theta)$; used to compute powers of complex numbers and to find n th roots. IB-MATHAA-1.5-1ANSWER	Mathematics: AA 1.4 The binomial theorem General term: $nCr \cdot 2^{(4-r)} \cdot (-x)^r$; $r=2$: $4C_2 \cdot 2^{2-1} = 6 \times 4 = 24$. IB-MATHAA-1.4-4ANSWER
Mathematics: AA 1.6 Proof by induction (HL) 1) Base case verification, 2) Inductive step: assume $P(k)$ true, prove $P(k+1)$ true, 3) Conclusion stating the principle of mathematical induction applies for all $n \geq n_0$. IB-MATHAA-1.6-1ANSWER	Mathematics: AA 1.5 Complex numbers (HL) Moduli multiply, arguments add: $z_1 z_2 = 6(\cos 70^\circ + i \sin 70^\circ)$. IB-MATHAA-1.5-4ANSWER	Mathematics: AA 1.5 Complex numbers (HL) B. $(2+i)(1-i) = 2-2i+i-i^2 = 2-i+1 = 3-i$. IB-MATHAA-1.5-3ANSWER



Mathematics: AA 1.6 Proof by induction (HL)	Mathematics: AA 1.6 Proof by induction (HL)	Mathematics: AA 1.6 Proof by induction (HL)
Check $n=1$: $2^1=2 > 1$, true, so the base case holds.	B. Induction applies to statements indexed by integers, typically all $n \geq$ some starting value.	Because checking finitely many cases does not prove the statement for all n ; the inductive step is essential to show truth propagates from any k to $k+1$, covering the infinite set of remaining cases.
IB-MATHAA-1.6-4ANSWER	IB-MATHAA-1.6-3ANSWER	IB-MATHAA-1.6-2ANSWER
Mathematics: AA 1.7 Systems of linear equations (HL)	Mathematics: AA 1.7 Systems of linear equations (HL)	Mathematics: AA 1.7 Systems of linear equations (HL)
B. Swapping rows reorders equations without altering the solution set; the other options can change or lose information.	Add or subtract a suitable multiple of one equation (row) to/from another so that the coefficient of the chosen variable becomes zero.	Unique solution (three planes meet at one point), no solution (inconsistent, e.g. parallel/skew planes with no common point), infinitely many solutions (planes intersect in a common line).
IB-MATHAA-1.7-3ANSWER	IB-MATHAA-1.7-2ANSWER	IB-MATHAA-1.7-1ANSWER
Mathematics: AA 2.1 Functions, domain and range	Mathematics: AA 2.1 Functions, domain and range	Mathematics: AA 1.7 Systems of linear equations (HL)
The expression under a square root (or even root) is negative, or a denominator equals zero — both make $f(x)$ undefined.	The vertical line test: if any vertical line crosses the graph more than once, the relation is not a function, since one input would map to more than one output.	Adding: $3x=6$, $x=2$, then $y=5-2=3$.
IB-MATHAA-2.1-2ANSWER	IB-MATHAA-2.1-1ANSWER	IB-MATHAA-1.7-4ANSWER
Mathematics: AA 2.2 Composite and inverse functions	Mathematics: AA 2.1 Functions, domain and range	Mathematics: AA 2.1 Functions, domain and range
They are reflections of each other in the line $y=x$.	Need $x+4 \geq 0$, so domain is $x \geq -4$.	A. Squares are never negative, and 0 is attained at $x=0$, so range is $f(x) \geq 0$.
IB-MATHAA-2.2-1ANSWER	IB-MATHAA-2.1-4ANSWER	IB-MATHAA-2.1-3ANSWER
Mathematics: AA 2.2 Composite and inverse functions	Mathematics: AA 2.2 Composite and inverse functions	Mathematics: AA 2.2 Composite and inverse functions
$g(1)=4$, $f(4)=8$.	B. Swapping and solving $y=x+5$ gives $f^{-1}(x)=x-5$.	If two different x -values give the same output, the inverse would have to map that one output value to two different inputs, which violates the definition of a function.
IB-MATHAA-2.2-4ANSWER	IB-MATHAA-2.2-3ANSWER	IB-MATHAA-2.2-2ANSWER
Mathematics: AA 2.3 Quadratic functions and the discriminant	Mathematics: AA 2.3 Quadratic functions and the discriminant	Mathematics: AA 2.3 Quadratic functions and the discriminant
C. $\Delta=16-20=-4<0$, so no real roots.	The vertex has x -coordinate $x=-b/(2a)$; substitute this back into $f(x)$ to find the y -coordinate.	$\Delta=b^2-4ac$. $\Delta>0$: two distinct real roots; $\Delta=0$: one repeated real root; $\Delta<0$: no real roots (complex conjugate pair).
IB-MATHAA-2.3-3ANSWER	IB-MATHAA-2.3-2ANSWER	IB-MATHAA-2.3-1ANSWER
Mathematics: AA 2.4 Transformations of graphs	Mathematics: AA 2.4 Transformations of graphs	Mathematics: AA 2.3 Quadratic functions and the discriminant
It compresses the graph vertically towards the x -axis by factor $ a $ and reflects it in the x -axis.	$y=f(x)+d$ translates the graph vertically by d (up if $d>0$). $y=f(x+c)$ translates it horizontally by $-c$ (i.e. c units left if $c>0$).	$x-(-2)/2=1$, $f(1)=1-2+4=3$. Vertex $(1,3)$.
IB-MATHAA-2.4-2ANSWER	IB-MATHAA-2.4-1ANSWER	IB-MATHAA-2.3-4ANSWER



Mathematics: AA 2.5 Exponential and logarithmic functions	Mathematics: AA 2.4 Transformations of graphs	Mathematics: AA 2.4 Transformations of graphs
Domain $x > 0$, range all real numbers, vertical asymptote $x = 0$ (no horizontal asymptote).	(1, 3), since the graph is translated down 2 units.	B. Multiplying the output by 3 stretches the graph vertically by factor 3.
IB-MATHAA-2.5-1ANSWER	IB-MATHAA-2.4-4ANSWER	IB-MATHAA-2.4-3ANSWER
Mathematics: AA 2.5 Exponential and logarithmic functions	Mathematics: AA 2.5 Exponential and logarithmic functions	Mathematics: AA 2.5 Exponential and logarithmic functions
As $x \rightarrow \infty$, $0.5^x \rightarrow 0$, so $y \rightarrow -4$: asymptote $y = -4$.	C. $4^0 + 1 = 1 + 1 = 2$.	Because $\ln(x)$ is defined as the inverse function of e^x , and the graph of any function's inverse is always the reflection of the original graph in $y = x$.
IB-MATHAA-2.5-4ANSWER	IB-MATHAA-2.5-3ANSWER	IB-MATHAA-2.5-2ANSWER
Mathematics: AA 2.6 Rational functions (HL)	Mathematics: AA 2.6 Rational functions (HL)	Mathematics: AA 2.6 Rational functions (HL)
B. Denominator zero: $2x - 6 = 0$, $x = 3$.	Because rewriting it as $A + B/(cx + d)$ ($B \neq 0$) shows the fraction term $B/(cx + d)$ is never exactly zero for finite x , so $y \neq A$ for any real x .	Vertical asymptote: solve $cx + d = 0$ for x . Horizontal asymptote: $y = a/c$, the ratio of the leading coefficients (from long division as $x \rightarrow \pm \infty$).
IB-MATHAA-2.6-3ANSWER	IB-MATHAA-2.6-2ANSWER	IB-MATHAA-2.6-1ANSWER
Mathematics: AA 2.7 Polynomial functions and factor theorem (HL)	Mathematics: AA 2.7 Polynomial functions and factor theorem (HL)	Mathematics: AA 2.6 Rational functions (HL)
The remainder theorem gives the remainder $P(k)$ when $P(x)$ is divided by $(x - k)$ for ANY k (which may be nonzero); the factor theorem is the special case where that remainder equals zero, showing $(x - k)$ divides $P(x)$ exactly.	$(x - k)$ is a factor of polynomial $P(x)$ if and only if $P(k) = 0$.	$y = 4/2 = 2$.
IB-MATHAA-2.7-2ANSWER	IB-MATHAA-2.7-1ANSWER	IB-MATHAA-2.6-4ANSWER
Mathematics: AA 3.1 Volume and surface area of 3D solids	Mathematics: AA 2.7 Polynomial functions and factor theorem (HL)	Mathematics: AA 2.7 Polynomial functions and factor theorem (HL)
Cone: $V = (1/3)\pi r^2 h$. Sphere: $V = (4/3)\pi r^3$.	By the remainder theorem, remainder = $P(2) = 8 - 4 + 1 = 5$.	C. Since $P(2) = 0$, by the factor theorem $(x - 2)$ is a factor.
IB-MATHAA-3.1-1ANSWER	IB-MATHAA-2.7-4ANSWER	IB-MATHAA-2.7-3ANSWER
Mathematics: AA 3.1 Volume and surface area of 3D solids	Mathematics: AA 3.1 Volume and surface area of 3D solids	Mathematics: AA 3.1 Volume and surface area of 3D solids
$V = 3 \times 4 \times 5 = 60 \text{ cm}^3$.	B. Curved surface area = $2\pi rh = 2\pi(2)(7) = 28\pi$.	The slant height l is the distance from the apex to the edge of the base circle, found using Pythagoras: $l = \sqrt{r^2 + h^2}$, where r is the base radius and h the vertical height.
IB-MATHAA-3.1-4ANSWER	IB-MATHAA-3.1-3ANSWER	IB-MATHAA-3.1-2ANSWER
Mathematics: AA 3.2 Right-angled trigonometry and bearings	Mathematics: AA 3.2 Right-angled trigonometry and bearings	Mathematics: AA 3.2 Right-angled trigonometry and bearings
B. They are equal, being alternate angles formed by the horizontal lines and the line of sight.	Bearings are measured clockwise from north, and always written as three digits (e.g. 045° , 302°), ranging from 000° to 360° .	$\sin \theta = \text{opposite/hypotenuse}$, $\cos \theta = \text{adjacent/hypotenuse}$, $\tan \theta = \text{opposite/adjacent}$.
IB-MATHAA-3.2-3ANSWER	IB-MATHAA-3.2-2ANSWER	IB-MATHAA-3.2-1ANSWER



Mathematics: AA 3.3 Non-right triangles: sine and cosine rules	Mathematics: AA 3.3 Non-right triangles: sine and cosine rules	Mathematics: AA 3.2 Right-angled trigonometry and bearings
It arises when given two sides and a non-included angle (SSA); after finding one possible angle via arcsin, check if 180° minus that angle also gives a valid triangle (angle sum <180° and consistent with given info).	Sine rule: $a/\sin A = b/\sin B = c/\sin C$. Cosine rule: $a^2 = b^2 + c^2 - 2bc \cdot \cos A$.	height = $6 \sin(60^\circ) = 6(0.8660) = 5.20$ m.
IB-MATHAA-3.3-2ANSWER	IB-MATHAA-3.3-1ANSWER	IB-MATHAA-3.2-4ANSWER
Mathematics: AA 3.4 The unit circle and radian measure	Mathematics: AA 3.3 Non-right triangles: sine and cosine rules	Mathematics: AA 3.3 Non-right triangles: sine and cosine rules
Arc length $s=r\theta$; sector area $A=(1/2)r^2\theta$.	$c^2=36+36-2(6)(6)\cos 60^\circ=72-72(0.5)=36$ $c=6$.	A. $\text{Area}=(1/2)(6)(8)\sin 90^\circ=(1/2)(48)(1)=24$.
IB-MATHAA-3.4-1ANSWER	IB-MATHAA-3.3-4ANSWER	IB-MATHAA-3.3-3ANSWER
Mathematics: AA 3.4 The unit circle and radian measure	Mathematics: AA 3.4 The unit circle and radian measure	Mathematics: AA 3.4 The unit circle and radian measure
$s=r\theta=5(\pi/2)=5\pi/2 \approx 7.85$ cm.	C. $\pi/3 \times 180/\pi = 60^\circ$.	For angle θ measured anticlockwise from the positive x-axis, $\cos\theta$ and $\sin\theta$ are respectively the x-coordinate and y-coordinate of the point where the terminal ray meets the unit circle.
IB-MATHAA-3.4-4ANSWER	IB-MATHAA-3.4-3ANSWER	IB-MATHAA-3.4-2ANSWER
Mathematics: AA 3.5 Trigonometric identities and equations	Mathematics: AA 3.5 Trigonometric identities and equations	Mathematics: AA 3.5 Trigonometric identities and equations
B. This is the Pythagorean identity, true for all x.	Because dividing the domain by 2 too early loses solutions; you must first find all values of $2x$ in the extended domain $[0, 2 \times \text{upper bound}]$, then divide each by 2 to get x, ensuring no solutions for x are missed.	$\sin^2\theta + \cos^2\theta = 1$. $\cos(2\theta) = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$ (all equivalent forms).
IB-MATHAA-3.5-3ANSWER	IB-MATHAA-3.5-2ANSWER	IB-MATHAA-3.5-1ANSWER
Mathematics: AA 3.6 Vectors and scalar/vector products (HL)	Mathematics: AA 3.6 Vectors and scalar/vector products (HL)	Mathematics: AA 3.5 Trigonometric identities and equations
$ a \times b $ equals the area of the parallelogram formed by a and b; the area of the triangle formed by those two vectors as sides is half of this, $(1/2) a \times b $.	$a \cdot b = a b \cos\theta$, where θ is the angle between the vectors.	$x = \pi/6$ or $x = \pi - \pi/6 = 5\pi/6$.
IB-MATHAA-3.6-2ANSWER	IB-MATHAA-3.6-1ANSWER	IB-MATHAA-3.5-4ANSWER
Mathematics: AA 3.7 Lines and planes in three dimensions (HL)	Mathematics: AA 3.6 Vectors and scalar/vector products (HL)	Mathematics: AA 3.6 Vectors and scalar/vector products (HL)
Line: $r = a + tb$, where a is a position vector to a known point on the line and b is a direction vector, t a parameter. Plane: $n \cdot r = n \cdot a$ (equivalently $ax+by+cz=d$), where $n=(a,b,c)$ is a normal vector to the plane and the point a lies on it.	$ v = \sqrt{9+16+0} = \sqrt{25} = 5$.	B. $a \cdot b = 1(0)+0(1)+0(0) = 0$, since they are perpendicular unit vectors.
IB-MATHAA-3.7-1ANSWER	IB-MATHAA-3.6-4ANSWER	IB-MATHAA-3.6-3ANSWER



Mathematics: AA 3.7 Lines and planes in three dimensions (HL) Substitute: $t+t+t=6 \rightarrow 3t=6 \rightarrow t=2$, point (2,2,2). IB-MATHAA-3.7-4ANSWER	Mathematics: AA 3.7 Lines and planes in three dimensions (HL) B. $n_1 \cdot n_2 = 0$, so the normals (and hence the planes) are perpendicular. IB-MATHAA-3.7-3ANSWER	Mathematics: AA 3.7 Lines and planes in three dimensions (HL) Find the angle ϕ between the direction vector and the normal using $\cos\phi = d \cdot n /(d n)$, then the angle between the line and the plane is $90^\circ - \phi$. IB-MATHAA-3.7-2ANSWER
Mathematics: AA 4.1 Sampling and data collection C. Simple random sampling gives each member an equal probability of being chosen. IB-MATHAA-4.1-3ANSWER	Mathematics: AA 4.1 Sampling and data collection Sampling bias occurs when a sample systematically over- or under-represents parts of the population, e.g. from using convenience sampling, which excludes anyone not easily accessible to the researcher. IB-MATHAA-4.1-2ANSWER	Mathematics: AA 4.1 Sampling and data collection The population is divided into distinct subgroups (strata), and a random sample is taken from each stratum proportional to its size, ensuring all subgroups are fairly represented. IB-MATHAA-4.1-1ANSWER
Mathematics: AA 4.2 Statistical measures and displays The mean is sensitive to extreme values since it uses every data value in its calculation, while the median depends only on the middle value(s) and is therefore resistant (robust) to outliers. IB-MATHAA-4.2-2ANSWER	Mathematics: AA 4.2 Statistical measures and displays $IQR = Q3 - Q1$. A value is typically considered an outlier if it is below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$. IB-MATHAA-4.2-1ANSWER	Mathematics: AA 4.1 Sampling and data collection $\text{Fraction} = 50/1000 = 1/20$. Women: $600/20 = 30$. IB-MATHAA-4.1-4ANSWER
Mathematics: AA 4.3 Correlation and linear regression It measures the strength and direction of the linear relationship between two variables; r ranges from -1 (perfect negative linear correlation) to +1 (perfect positive linear correlation), with 0 meaning no linear correlation. IB-MATHAA-4.3-1ANSWER	Mathematics: AA 4.2 Statistical measures and displays $\text{Range} = 14 - 3 = 11$. Median = 8; $Q1 = 5$, $Q3 = 11$; $IQR = 11 - 5 = 6$. IB-MATHAA-4.2-4ANSWER	Mathematics: AA 4.2 Statistical measures and displays C. The line inside the box of a box plot represents the median. IB-MATHAA-4.2-3ANSWER
Mathematics: AA 4.3 Correlation and linear regression $y = 3(4) + 2 = 14$. IB-MATHAA-4.3-4ANSWER	Mathematics: AA 4.3 Correlation and linear regression C. r closest to 0 indicates the weakest linear relationship. IB-MATHAA-4.3-3ANSWER	Mathematics: AA 4.3 Correlation and linear regression Because the linear model was fitted only to data within a certain range, there is no evidence the same linear relationship continues to hold outside that range, so predictions far beyond it may be inaccurate. IB-MATHAA-4.3-2ANSWER
Mathematics: AA 4.4 Probability, Venn and tree diagrams A. Since $P(A) \times P(B) = 0.18 = P(A \cap B)$, the independence condition holds. IB-MATHAA-4.4-3ANSWER	Mathematics: AA 4.4 Probability, Venn and tree diagrams A and B are independent if $P(A \cap B) = P(A) \times P(B)$, equivalently $P(A B) = P(A)$ (knowing B occurred does not change the probability of A). IB-MATHAA-4.4-2ANSWER	Mathematics: AA 4.4 Probability, Venn and tree diagrams $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. If A and B are mutually exclusive, $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$. IB-MATHAA-4.4-1ANSWER



Mathematics: AA 4.5 Discrete random variables and the binomial distribution A fixed number n of independent trials, each with only two outcomes (success/failure), and a constant probability p of success on each trial. IB-MATHAA-4.5-2ANSWER	Mathematics: AA 4.5 Discrete random variables and the binomial distribution $E(X)=np$, $\text{Var}(X)=np(1-p)$. IB-MATHAA-4.5-1ANSWER	Mathematics: AA 4.4 Probability, Venn and tree diagrams $P(A \cup B)=0.4+0.5-0=0.9$. IB-MATHAA-4.4-4ANSWER
Mathematics: AA 4.6 The normal distribution $z = (x-\mu)/\sigma$, where μ is the mean and σ the standard deviation of X . IB-MATHAA-4.6-1ANSWER	Mathematics: AA 4.5 Discrete random variables and the binomial distribution $E(X)=5(0.5)=2.5$. $\text{Var}(X)=5(0.5)(0.5)=1.25$. IB-MATHAA-4.5-4ANSWER	Mathematics: AA 4.5 Discrete random variables and the binomial distribution C. The total probability across all possible outcomes must sum to 1. IB-MATHAA-4.5-3ANSWER
Mathematics: AA 4.6 The normal distribution $z=(25-20)/5=1$. IB-MATHAA-4.6-4ANSWER	Mathematics: AA 4.6 The normal distribution C. The normal distribution is symmetric about its mean, so exactly half the probability lies below μ . IB-MATHAA-4.6-3ANSWER	Mathematics: AA 4.6 The normal distribution Approximately 68% within 1σ , 95% within 2σ , and 99.7% within 3σ of the mean. IB-MATHAA-4.6-2ANSWER
Mathematics: AA 4.7 Bayes' theorem (HL) B. $P(A)$ is the prior probability, updated to the posterior $P(A B)$ after observing evidence B. IB-MATHAA-4.7-3ANSWER	Mathematics: AA 4.7 Bayes' theorem (HL) It computes the denominator $P(B)$ by summing over all ways B can occur: $P(B) = P(B A)P(A) + P(B A')P(A')$, needed since $P(B)$ is usually not given directly. IB-MATHAA-4.7-2ANSWER	Mathematics: AA 4.7 Bayes' theorem (HL) $P(A B) = P(B A) \cdot P(A) / P(B)$. IB-MATHAA-4.7-1ANSWER
Mathematics: AA 5.1 Limits and the derivative from first principles The gradient of the tangent line to the curve $y=f(x)$ at the point x , i.e. the instantaneous rate of change of f at that point. IB-MATHAA-5.1-2ANSWER	Mathematics: AA 5.1 Limits and the derivative from first principles $f'(x) = \lim_{h \rightarrow 0} [f(x+h)-f(x)]/h$. IB-MATHAA-5.1-1ANSWER	Mathematics: AA 4.7 Bayes' theorem (HL) $P(A B)=P(B A)P(A)/P(B)=(0.8 \times 0.25)/0.4=0.2$. IB-MATHAA-4.7-4ANSWER
Mathematics: AA 5.2 Differentiation rules and tangents/normals Product rule: $d/dx(uv)=u'v+uv'$. Quotient rule: $d/dx(u/v)=(u'v-uv')/v^2$. IB-MATHAA-5.2-1ANSWER	Mathematics: AA 5.1 Limits and the derivative from first principles $[4(x+h)-4x]/h = 4h/h = 4$, so $f'(x)=4$. IB-MATHAA-5.1-4ANSWER	Mathematics: AA 5.1 Limits and the derivative from first principles B. $[f(x+h)-f(x)]/h = [5-5]/h = 0$ for all $h \neq 0$, so the limit is 0. IB-MATHAA-5.1-3ANSWER
Mathematics: AA 5.2 Differentiation rules and tangents/normals $dy/dx=2x+3$, at $x=2$: $4+3=7$. IB-MATHAA-5.2-4ANSWER	Mathematics: AA 5.2 Differentiation rules and tangents/normals C. Power rule: $d/dx(x^5)=5x^4$. IB-MATHAA-5.2-3ANSWER	Mathematics: AA 5.2 Differentiation rules and tangents/normals The normal is perpendicular to the tangent, so if the tangent gradient is m ($m \neq 0$), the normal gradient is $-1/m$. IB-MATHAA-5.2-2ANSWER



Mathematics: AA 5.3 Optimisation and related rates	Mathematics: AA 5.3 Optimisation and related rates	Mathematics: AA 5.3 Optimisation and related rates
A. $f'(x)=2x-6=0 \rightarrow x=3$ (and $f''(x)=2>0$ confirms minimum).	It links the rate of change of one quantity to another through a shared variable, e.g. $dV/dt = (dV/dr) \times (dr/dt)$, allowing you to find an unknown rate from a known one and a known relationship between the quantities.	If $f''(a)>0$, $x=a$ is a local minimum; if $f''(a)<0$, $x=a$ is a local maximum; if $f''(a)=0$, the test is inconclusive and another method (e.g. sign change of f') is needed.
IB-MATHAA-5.3-3ANSWER	IB-MATHAA-5.3-2ANSWER	IB-MATHAA-5.3-1ANSWER
Mathematics: AA 5.4 Kinematics using calculus	Mathematics: AA 5.4 Kinematics using calculus	Mathematics: AA 5.3 Optimisation and related rates
Set the velocity function $v(t)=0$ and solve for t ; these are the instants where the particle's motion momentarily stops (though it may not be a permanent stop).	$v(t)=ds/dt$, $a(t)=dv/dt=d^2s/dt^2$. Conversely, $v=\int a \, dt$ and $s=\int v \, dt$.	$f'(x)=2x-4=0 \rightarrow x=2$. $f''(x)=2>0$, so $x=2$ is a minimum.
IB-MATHAA-5.4-2ANSWER	IB-MATHAA-5.4-1ANSWER	IB-MATHAA-5.3-4ANSWER
Mathematics: AA 5.5 Integration and areas under curves	Mathematics: AA 5.4 Kinematics using calculus	Mathematics: AA 5.4 Kinematics using calculus
$\int x^n \, dx = x^{(n+1)}/(n+1)+C$ ($n \neq -1$). Also $\int e^x \, dx = e^x + C$ and $\int 1/x \, dx = \ln x + C$.	$v(t)=2t-4$, at $t=3$: $6-4=2$ m/s.	B. $v(t)=\int 4 \, dt=4t+C$; $v(0)=2$ gives $C=2$, so $v(t)=4t+2$.
IB-MATHAA-5.5-1ANSWER	IB-MATHAA-5.4-4ANSWER	IB-MATHAA-5.4-3ANSWER
Mathematics: AA 5.5 Integration and areas under curves	Mathematics: AA 5.5 Integration and areas under curves	Mathematics: AA 5.5 Integration and areas under curves
x^3-2x+C .	C. $\int 2x \, dx = x^2$, evaluated: $9-1=8$.	The area formula $\int (f(x)-g(x))dx$ requires $f(x) \geq g(x)$ on the interval to give a positive area; subtracting in the wrong order gives a negative value that must otherwise be corrected by taking the absolute value or swapping the functions.
IB-MATHAA-5.5-4ANSWER	IB-MATHAA-5.5-3ANSWER	IB-MATHAA-5.5-2ANSWER
Mathematics: AA 5.6 Volumes of revolution (HL)	Mathematics: AA 5.6 Volumes of revolution (HL)	Mathematics: AA 5.6 Volumes of revolution (HL)
B. $V=\pi \int [0 \text{ to } 5] 4 \, dx = \pi(4 \times 5) = 20\pi$.	Because the solid is modelled as a stack of thin circular discs of radius y , and the area of a disc is πy^2 , so summing (integrating) πy^2 over the thickness dx gives the total volume.	$V = \pi \int [a \text{ to } b] y^2 \, dx = \pi \int [a \text{ to } b] [f(x)]^2 \, dx$.
IB-MATHAA-5.6-3ANSWER	IB-MATHAA-5.6-2ANSWER	IB-MATHAA-5.6-1ANSWER
Mathematics: AA 5.7 Further differentiation: implicit and higher derivatives (HL)	Mathematics: AA 5.7 Further differentiation: implicit and higher derivatives (HL)	Mathematics: AA 5.6 Volumes of revolution (HL)
Because such equations cannot easily be rearranged into the explicit form $y=f(x)$ (or would require splitting into multiple branches), so differentiating both sides directly while treating y as an implicit function of x is more efficient.	Apply the chain rule, treating y as a function of x : $d/dx(y^2)=2y \cdot dy/dx$. For a product term like xy , use the product rule: $d/dx(xy)=y+x \cdot dy/dx$.	$V = \pi \int [0 \text{ to } 1] x^4 \, dx$.
IB-MATHAA-5.7-2ANSWER	IB-MATHAA-5.7-1ANSWER	IB-MATHAA-5.6-4ANSWER

**Mathematics: AA**

5.8 Further integration: substitution and by parts (HL)

$$\int u \, dv = uv - \int v \, du.$$

IB-MATHAA-5.8-1

ANSWER

Mathematics: AA

5.7 Further differentiation: implicit and higher derivatives (HL)

$$2x - dy/dx = 0, \text{ so } dy/dx = 2x.$$

IB-MATHAA-5.7-4

ANSWER

Mathematics: AA

5.7 Further differentiation: implicit and higher derivatives (HL)

$$\text{A. Differentiating: } 1 + 2y(dy/dx) = 0, \\ \text{so } dy/dx = -1/(2y).$$

IB-MATHAA-5.7-3

ANSWER

Mathematics: AA

5.8 Further integration: substitution and by parts (HL)

Let $u = x^2$, $du = 2x \, dx$. Integral becomes $\int \cos(u) du = \sin(u) + C = \sin(x^2) + C$.

IB-MATHAA-5.8-4

ANSWER

Mathematics: AA

5.8 Further integration: substitution and by parts (HL)

B. Since $du = 3x^2 dx$ exactly matches the rest of the integrand, $u = x^3 + 1$ is the natural substitution.

IB-MATHAA-5.8-3

ANSWER

Mathematics: AA

5.8 Further integration: substitution and by parts (HL)

You must convert dx into an expression in terms of du (using $du = g'(x)dx$), and ideally rewrite the entire integral purely in terms of u before integrating.

IB-MATHAA-5.8-2

ANSWER

Mathematics: AA

5.9 Differential equations and Maclaurin series (HL)

B. $\sin(x) = x - x^3/6 + x^5/120 - \dots$, so the coefficient of x^3 is $-1/6$.

IB-MATHAA-5.9-3

ANSWER

Mathematics: AA

5.9 Differential equations and Maclaurin series (HL)

$f(x) = f(0) + f'(0)x + f''(0)x^2/2! + f'''(0)x^3/3! + \dots$, using successive derivatives of f evaluated at $x=0$.

IB-MATHAA-5.9-2

ANSWER

Mathematics: AA

5.9 Differential equations and Maclaurin series (HL)

Rearrange to $(1/g(y))dy = f(x)dx$, integrate both sides separately, then apply any given initial condition to determine the constant of integration.

IB-MATHAA-5.9-1

ANSWER

Mathematics: AA

5.9 Differential equations and Maclaurin series (HL)

$(1/y)dy = 2dx \rightarrow \ln|y| = 2x + C \rightarrow y = Ae^{(2x)}$; $y(0) = 5$ gives $A = 5$, so $y = 5e^{(2x)}$.

IB-MATHAA-5.9-4

ANSWER